Project 14/20 — AI Interview Preparation Platform

A personalized AI mock-interview system for students and job seekers

1. What does this project do?

The platform helps users prepare for technical, HR, and role-specific interviews. A user uploads a resume, selects a target role, and the system generates a personalized interview session.

• Analyze the user's resume and identify relevant skills and projects.

• Generate interview questions for a selected role.

• Conduct a mock interview through text or voice.

• Evaluate answers using an LLM and structured scoring criteria.

• Provide feedback and a personalized practice plan.

• Track progress across multiple interview sessions.

2. Example user flow

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Upload resume → Select job role → Generate interview → Answer questions → AI evaluates answers → Receive report → Practice weak areas → Repeat.
```

3. Example

Target role: Python Backend Developer
Resume: FastAPI + PostgreSQL + Docker projects
AI question: Explain how dependency injection works in FastAPI.
User answer: The candidate records or types an answer.
Feedback: The system identifies missing concepts, gives a model explanation, and suggests a follow-up question.

4. How can it make money?

Business model	Example
Freemium	A limited number of mock interviews for free.
Subscription	■99–■499/month for more interviews and advanced reports.
One-time package	Interview-preparation package for a specific role.
B2B / colleges	License the platform to training or placement programs.

Illustrative example: 200 users paying ■199/month would produce ■39,800/month in revenue before AI, hosting, payment, and other costs. Actual income depends on acquiring and retaining paying customers.

5. Core architecture

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User → React interface → FastAPI backend → Resume/document parser → LLM interview engine → Evaluation service → PostgreSQL → Progress dashboard
```

6. Suggested technology stack

Layer	Technology
Frontend	React / Next.js
Backend	Python + FastAPI
Database	PostgreSQL
AI	LLM API or local model
Document processing	PDF/text extraction
Voice (optional)	Speech-to-Text + Text-to-Speech
Deployment	Docker + cloud/VPS

7. AI evaluation system

Instead of asking the model for an arbitrary score, define a consistent rubric. For example: correctness, completeness, clarity, relevance, and communication. The AI can return structured JSON containing scores, evidence from the answer, missing concepts, and improvement suggestions.

8. Development roadmap

• Version 1: Resume upload + role selection + generated questions.

• Version 2: Text-based mock interview + feedback report.

• Version 3: User accounts + interview history + progress dashboard.

• Version 4: Voice interview with speech-to-text and text-to-speech.

• Version 5: Role-specific question banks, analytics, and subscription payments.

• Version 6: College/company dashboard for bulk interview practice.

9. Skills you will learn

FastAPI, PostgreSQL, authentication, LLM integration, structured outputs, prompt engineering, PDF processing, speech APIs, background jobs, analytics, payments, Docker, and cloud deployment.

10. Difficulty and build time

Difficulty: Intermediate → Advanced
Initial MVP: approximately 2–3 weeks
Full voice-enabled SaaS: approximately 4–6 weeks
Portfolio value: Strong combination of AI + backend + product engineering.

11. Important product considerations

Treat AI feedback as practice guidance rather than an authoritative hiring decision. Give users the underlying feedback and examples so they can judge and improve their own answers. If storing resumes or interview recordings, implement appropriate access controls, retention rules, and data deletion mechanisms.

12. Portfolio-ready project title

AI Interview Coach — Personalized Mock Interview & Feedback Platform
A full-stack AI application that analyzes resumes, generates role-specific interviews, conducts text/voice mock interviews, evaluates responses, and tracks preparation progress.